

Assignment 2: Projectile Motion Continued

Assignment for COSC 1200 – Object-Oriented Programming 1

Prior to attempting to solve this problem, you should have viewed the Content, including assigned readings for weeks 1 through 7.

Use of uncited AI-generated code or code from help websites may result in escalation as an academic integrity alert at the professor's discretion.

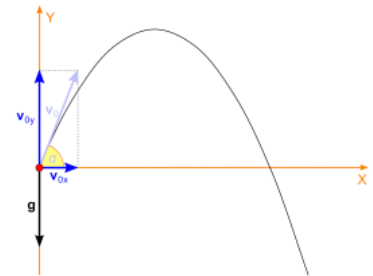
Use of other reference material is encouraged. **Proof of understanding** and **proper referencing** are strongly encouraged for code copied from any other source.

General Requirements

1. This assignment is to be completed **individually**. Group submissions will not be considered.
2. Submit your solution to the appropriate assignment folder on DC Connect **by the due date provided**.
3. Your instructor will assign a grade and post feedback on your submission to DC Connect.

Background Information

This program is a game that simulates two players firing at each other using something like artillery. This will be based on a calculation of projectile motion neglecting air resistance. The provided diagram represents a projectile's motion under the influence of gravity.



Formulas will be used to calculate the projectile's position at a given moment in time. Its position at any point in time (t) is calculated separately for the vertical and horizontal directions.

Horizontal – the distance travelled horizontally at time t can be calculated by:

$$d_x = (v_0 \times \cos\theta_0) t$$

Vertical – the vertical distance travelled at time t can be calculated by:

$$d_y = (v_0 \times \sin\theta_0) t - \frac{1}{2}gt^2$$

Where:

- v_0 – represents the value of the projectiles initial speed at time 0
- θ_0 – represents the initial angle of the projectile relative to the horizontal surface at time 0
- t – represents the time that has passed in seconds
- g – represents the gravitation force constant (typically -9.81 m/s^2)

Program Requirements

The program should start by requesting the names for each player. It will refer to the players by their names for the rest of the program. Initialize each player's score to 0.

In each individual game, both players will begin at a random position (an x-coordinate) between 0 and 400.

Create the following functions:

- `getPower()`, returning a double value: Prompt the user for their “power” or velocity value, which will replace v_0 (initial velocity) in the provided formula. This can be any number between 1 and 1000. Collect validated input based on this.
- `getAngle()`, returning a double value: Prompt the user for θ_0 – their initial projectile angle. This should be between 0 and 180. It will return a validated input value.
- `getImpact()`, returning a double value: This will call both `getPower()` and `getAngle()`, and use those values to determine and return the x-coordinate where the shot lands. Specifically, this means:

Assignment 2: Projectile Motion Continued

Assignment for COSC 1200 – Object-Oriented Programming 1

1. Use $d_y = (v_0 \times \sin\theta_0) - \frac{1}{2}gt^2$ to find a value for $t > 0$ (time after firing) when $y \leq 0$ (the shot has landed). (Some helpful tips related to the math will be discussed in class.)
 2. Once you have that value for t (time after firing), use $x = (v_0 \times \cos\theta_0) \times t + \text{startingX}$ to see the x-coordinate where the shot has landed.
- `runGame()`, returning a player number: This will initialize the player positions to random values between 0 and 120, and then repeatedly call `getImpact()` starting with the first player. After each `getImpact()`, it will report the x-coordinate where that player's shot landed. It will return the player number of the winner, once someone's `getImpact()` result is within < 1 of the target's location.

Additionally, the program should use a while loop to allow the user to indicate whether they would like to play another game. The program only ends when the user says, "no" to this prompt.

Other Requirements

- Include a comment header with your name, the date, and a short description in each human-generated code file.
- **Follow the style guide!** Use descriptive names and sensible data types for variables, constants, etc. that follow the style guide's naming conventions. Use appropriate spacing, indenting, and make sure braces `{ }` are located where they should be.
- Code must be commented in line with the style guide and your professor's expectations.
- Output messages must be meaningful. Write the output as if the user has not read these instructions.
- Any assets or code obtained from, or based closely on, external sources must be cited correctly, either in line with your professor's expectations or using APA format.

Submission and Grading

To earn full marks on this assignment, you must do *all* of the following:

- Push the source code files to GitHub prior to the due date.
 - Add your professor as a collaborator on the GitHub repository that includes the source code.
 - If you worked on a different version of this assignment previously, you may continue using the same repository.
- Submit the source code files to DC Connect prior to the due date.
 - Include your GitHub repository URL in the submission notes.
- Demonstrate the code file running one-on-one with your professor in the week following the due date. Your professor may allow a video or remote demonstration, or they may require you to attend class for this.

If any part of this is beyond the due date, the standard late penalty as noted in our Course Outline will apply.

The rubric for the grading of this assignment should be available on DC Connect.

Academic Integrity

By submitting this assignment to DC Connect for evaluation, you agree to the following terms:

- ☐ This work is your original work, with the majority its contents originally created by yourself.
- ☐ Any aspects of this work that were created by or based heavily on reference to another person's work, a website, a video, or a program (including generative AI systems) are cited with suitable identification to observe the original source.

Assignment 2: Projectile Motion Continued

Assignment for COSC 1200 – Object-Oriented Programming 1

- ☐ You did not transmit this file to anyone else's computer, nor receive any part of this work from anyone else's computer.

Failure to observe these points will leave you subject to Durham College's [Academic Integrity Policy](#), with a minimum penalty of a grade of 0 on the assignment and an academic integrity alert submitted to the faculty office.